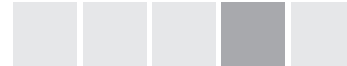


Involving Customers with Iterative Design

4



3 The primary principle of customer-centered design is *know your customers*, as discussed in Chapter 3 (Knowing Your Customers: Principles and Techniques). Now you will learn three related principles: (1) keep the customers involved, (2) conduct rapid prototyping, and (3) evaluate your designs. Omitting any of these principles from your design processes is a major risk. Studies by the Standish Group International have attributed many of information technology's frequent project failures to a lack of end-customer input.

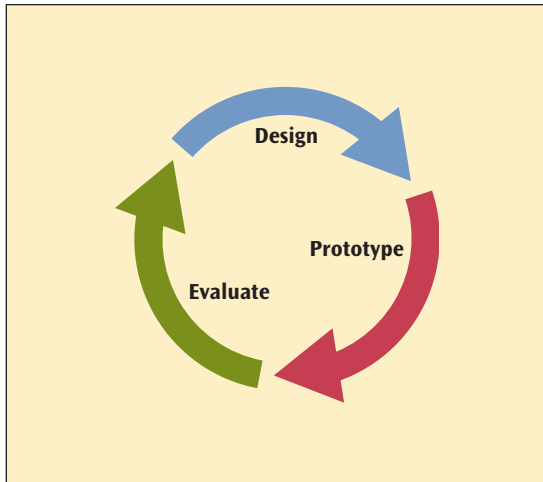
Iterative design addresses this problem by calling for setting measurable goals and repeatedly refining and testing design prototypes with customers until the final design meets or surpasses those goals. Your goals can be high-level and strategic, such as increased customer satisfaction or increased sales. They can also be short-term and tactical, such as reduced time to find items or fewer mouse clicks to check out and complete a purchase.

4.1 The Iterative Design Process

In **iterative design**, a new or existing design is reworked until it fits the needs of customers. This process is widely considered to be a valuable technique for designing interfaces, but at a more fundamental level iterative design acknowledges that *no design team is perfect*.

Teams are made of people, and people do not always know the right answers. It is difficult to have complete information about the needs of customers and clients. Iterative design compensates for these shortcomings, letting you continually improve a design. The key to iterative design is to quickly create design prototypes that are good enough to provide feedback but flexible enough for significant changes to be made

4 Involving Customers with Iterative Design



4.1

Figure 4.1

Iterative design is an ongoing cycle composed of three steps: design, prototype, and evaluate.

down the line. The goal is to conduct as much iteration as possible to solve as many problems as possible early, when they are inexpensive to fix.

The iterative design process has three steps (see Figure 4.1):

- 1. Design.** In this step, teams consider business goals and customer needs, setting measurable goals and developing design concepts.
- 2. Prototype.** In this step, teams develop artifacts as basic as scenarios and storyboards, and as complex as running Web sites, that illustrate how the site will accomplish the goals outlined in step 1.
- 3. Evaluate.** In this step, teams assess the prototypes developed in step 2 to see if they meet the desired goals.

The results of this process are used to inform design in the next iteration, and the entire process repeats until the goals have been met. Sometimes, however, scheduling and budgeting constraints force work to begin on the final production site before all the goals have been met.

Use design patterns to help you move quickly through each step of iterative design. During the initial design stage, use the information you learned about your customers, the genre patterns, and other high-level patterns to rough out the basic features that your site will need. Use the more detailed, low-level patterns to help you storyboard and prototype Web pages for specific scenarios. After evaluating your prototypes with

customers, clients, and other team members, use the patterns again to find solutions to the particular problems they encountered. Use these solutions in the next design iteration.

Before working on the final production site, go through this iterative cycle several times in the early stages of design and *make simple prototypes instead of full-fledged sites*. Real Web sites can take several weeks or months to implement, and by the time a site is completed and ready for evaluation, it might already be too expensive and time-consuming to fix any problems.

In contrast, prototypes can be created in just a few hours or days. The prototypes will not have all the features of a finished site, and many features will be faked, but they will be real enough to give customers a flavor of what the final site will be like. In this way you can get a lot of feedback about what works and what doesn't. (We talk more about how to create prototypes in Section 4.4, Rapid Prototyping.)

4.2 Reasons to Use Iterative Design

There are three main reasons to use an iterative design process:

1. It will help you find problems while they're still inexpensive and easy to fix.
2. It ensures that the site you're building will have the features that your customers need.
3. It ensures that you're building those features in a way that your customers can use.

Fixing Errors While They're Still Inexpensive and Easy to Fix

Why is it important to fix errors as early as possible? It has been well documented in many disciplines that fixing errors in the later phases of design can be expensive. The famous architect Frank Lloyd Wright¹ said it best: "An architect's most useful tools are an eraser at the drafting board, and a wrecking bar at the site." Clearly, using the wrecking bar and rebuilding is far more expensive than simply erasing and redesigning. In the realm of software development, a general rule of thumb is that errors cost about *ten times* more effort and money to fix late in the process than if they are caught in an earlier phase. Watts Humphrey and others in the

¹ Quoted in D. A. Aaker, *Brand Portfolio Strategy: Creating Relevance, Differentiation, Energy, Leverage, and Clarity* (New York: Free Press, 2004).

4 Involving Customers with Iterative Design

field of software engineering have even documented costs on the order of a hundred to a thousand times more effort and money to fix problems after deployment.

Why does it cost so much to make downstream changes? There are three reasons: (1) All the deliverables generated in later phases of design have to be made consistent with the proposed changes. (2) Sometimes one change forces other changes to be made so that everything will work correctly. (3) Most importantly, anything that causes a change in the software source code and HTML is expensive. Think about the time it takes to change a simple sketch versus the time it takes to rewrite the code that implements the ideas in the sketch.

Suppose that you already have a Web site but want to add LOCATION BREAD CRUMBS (K6), tiny markers like **Home > About > History**, at the top of each page so that your customers always know where they are in the site. Not only does this change have to be made on every page, but you also have to redesign the overall page layout to accommodate that change. Style sheets and other template mechanisms may make the technical part of this change much easier, but you will still have to make significant design and layout changes so that it all makes sense.

Now this is just a trivial change. Imagine the kind of effort it would take to make more involved changes, such as altering the way information is organized on the site or adding a significant feature that requires new code. Add these costs to what was spent in the first place, and you'll see why downstream changes are expensive.

Building the Right Site, and Building the Site Right

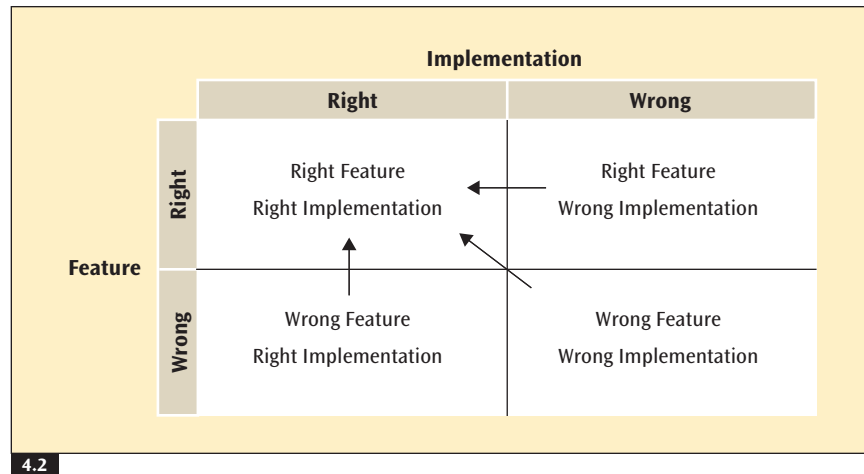
What kinds of mistakes are made on Web sites? Figure 4.2 categorizes the problems by feature and implementation. The top right-hand quadrant shows the right feature but the wrong implementation. For example, a SHOPPING CART (F3) is definitely the right feature for an e-commerce site, but it could have implementation problems that make it hard for customers to check out and finalize purchases. Iterative design and testing will help you discover these types of problems.

The bottom left-hand quadrant of Figure 4.2 shows the wrong feature but the right implementation. For example, providing extremely sophisticated search features for power users is not very useful if most of your customers cannot even understand the basic search capabilities.

The bottom right-hand quadrant shows the wrong feature and the wrong implementation—a design that is not useful *and* does not work correctly.

Figure 4.2

Iterative design helps design teams find the features that customers need (the “right” features) and make sure that those features are implemented correctly (the “right” implementation).



4.2

- 5 A good design process, like the one presented in Chapter 5 (Processes for Developing Customer-Centered Sites), should filter out most of these sorts of problems.

Ultimately, you want to be in the top left-hand quadrant, with the right features and the right implementations of those features. Iterative design helps push you toward the top left by getting constant feedback from customers about features and their implementation.

To summarize, design mistakes are costly, and it pays off in the long run to find them as early as possible in the development cycle. This is why rapid iterative design is so important in the early phases: it helps teams find and root out as many problems as possible *before* the site is deployed.

4.3 Designing with Goals and Principles in Mind

- 3 The design process is driven by both goals and principles. Goals come directly from analyzing your business and customer needs [see Chapter 3 (Knowing Your Customers: Principles and Techniques)]. Design principles come from research in human–computer interaction, as well as graphic design. Following these principles as you iterate on your design will help you reach your goals. Most of these goals and principles will apply whether you’re working on your site’s information architecture, navigation design, or graphic design.

Setting Measurable Design Goals

What does it mean to “get the site right”? This is where business, usability, and customer experience goals come into play. Some possible design goals include the following:

- Faster task completion
- Successful completion of more tasks
- Greater ease of learning
- Commission of fewer errors
- Abandonment of fewer shopping carts
- Greater pleasure or satisfaction
- More fun
- Increased visitor-to-customer conversion rate
- Increased customer repeat visits
- Increased revenue

Studies that we’ve conducted show that achieving many of the usability and customer experience goals, such as giving a more satisfying experience, have a direct impact on achieving business-related goals such as increased customer repeat visits.

The key to achieving all of these goals is testing and measuring. The rest of this section outlines examples of tests and measures that you can undertake to assess your progress toward your design goals.

For example, suppose that you want to know how long it takes to complete common tasks, such as checking out and finalizing a purchase. In this case, task completion time is an important metric. You could recruit representative customers to try out the site to measure the time it takes to complete these tasks, time how long it takes them to finish each task, see if they complete their tasks successfully, observe where they have problems, and determine whether there’s any commonality among the problems. Are the problems occurring in the same places? Are they navigation errors? Are they problems with the search feature? See if an existing design pattern from this book can help. Implement the necessary changes and retest to see if you solved the problems.

Suppose your server logs indicate that customers who start to post messages in the community section of your site have a low rate of completing the postings. A follow-on usability study confirms that customers are distracted in the middle of the posting process and tend to follow some tangential links, never to return. This finding might indicate a good place to apply a PROCESS FUNNEL (H1), which will help your customers go through the steps of completing their initial task: posting a message.

H1

Another approach to achieving your design goals is to compare two different prototypes to see which one works best. You might find that with one of the designs, customers are 20 percent faster on average than with the other design. A similar approach is to compare the new site design to the existing design or the design of one of your competitors' sites.

Alternatively, you might want to know how quickly customers get up to speed using the sites. Obtaining this information can be harder than it sounds. You could measure how many hours of use customers need on average to complete a particular set of tasks successfully. Or you could measure how many tasks customers complete in a certain amount of time. Both of these forms of ease-of-learning measures can be used to compare your site designs to one another and to the competition.

Another option is to measure and compare the average number of errors made on each site. If your customers make errors and become confused, they will not be able to achieve their goals on the site. For example, they will not be able to buy your products. To measure errors, first define them. Does clicking on the browser's **Back** button indicate an error? For example, suppose a customer is purchasing five cartridges of printer toner online. As he finishes the checkout process and fills out a form with his credit card number and address, he notices that he forgot to change the quantity on the order form on the previous page. The customer has to go back to correct the mistake. Is simply going back an error? When he goes forward again, the billing information might be lost. Is losing the billing information an error? Probably, but you have to define what you're measuring first.

Sometimes you can find issues with specific pieces of your site that do not necessarily indicate an interface problem but might indicate a general problem with your content or policies. For example, many sites measure the rate of shopping cart abandonment before checkout. The checkout process might be too long [this interface problem is addressed in QUICK-FLOW CHECKOUT (F1)], but there are several alternative explanations that may be based on your content. Customers might abandon their shopping carts if they cannot find all the products that they want to purchase. Another explanation is that your prices are higher than a competitor's, and people are just there to comparison-shop. A fourth explanation could be that customers are surprised at how high the sales tax or shipping and handling costs are and decide to leave. At any rate, the metric of shopping cart abandonment is certainly tied to your business's revenue numbers.

F1

The key to any of the numerical measurements discussed so far is to make sure that you're looking at numbers that make sense. This is where statistics and research design come into play. As we mentioned in Chapter 3 (Knowing Your Customers: Principles and Techniques), when designing surveys and usability studies you want to ensure that your numerical results are both reliable and valid. Be certain you have enough research participants to get statistically significant results. Often 20 to 50 participants might be required, depending on the variability in their background and performance.

You might also want to look at more subjective metrics. Do your customers find your site more pleasing or satisfying than the last version of the site, or your competitors' sites? Is your site fun? You can measure responses to questions such as these on an ongoing basis to understand how your site and your customers' opinions change over time. This information will help you know when you need to conduct more in-depth research. Again, these numbers can be tied directly to your bottom-line revenue or profit.

You can measure these subjective issues with surveys. Online surveys are easy to create and can be sent to a representative sample of your customers, or they can be made available on one of your Web pages. Target additional surveys for specific pages, or for times when customers take specific actions. Would you like to ask visitors why they abandoned their shopping carts? You can ask them with a survey, right when it happens!

The design goals that we've talked about so far are only an approximation of the higher-level client and business goals you might have. For example, shorter task completion time and less shopping cart abandonment are important metrics to work toward, but the business goal for an e-commerce site is to increase revenue. The problem is that you simply cannot gather this kind of metric using prototypes. Still, prototypes are good enough to provide useful feedback that will bring you closer to the overall goals.

Adhering to Design Principles

Design goals represent the destination you want to reach when you're finished building a Web site. Design *principles* guide you to that destination. The patterns in this book were guided by design principles, as well as by observation of what has worked well for customers. Here we present some basic design principles that you can use to tailor the patterns to your particular situation. Entire books could be written about these principles. The seven that we present are based on those from some of the most respected sources, specifically Ben Shneiderman's eight golden rules of

interface design, Jakob Nielsen's ten heuristics, and Edward Tufte's musings on information presentation:

1. Be consistent throughout. Consistency applies across several dimensions. Use a consistent sequence of actions to carry out similar tasks. Pages should have consistent color, layout, and fonts, implemented manually or by the use of STYLE SHEETS (D11). For example, your NAVIGATION BAR (K2) should be in the same place on every page, and ACTION BUTTONS (K4) that do the same thing should appear in the same general location across different parts of the site. Use identical terms in different places across the site. Make your site consistent with the real world: follow real-world conventions, and use FAMILIAR LANGUAGE (K11), sticking to terms that your customers will understand rather than technical jargon.

2. Offer informative feedback. Make the status of the system visible, and keep your customers informed about what's going on. For example, this is the principle behind SECURE CONNECTIONS (E6), which lets customers know whether the information that they're about to send over the Internet will be safe, and the PROGRESS BAR (H13), which indicates to the customer that a Web site is still functioning.

3. Rely on recognition over recall. Short-term memory is the key limitation in human cognition. Reducing the short-term memory load is easy if people can *recognize* what they need to know from visible objects, actions, options, and directions. The memory load is much higher if they need to *recall* this information from memory with no visual aids. This is why a visual user interface like that of the Macintosh or Windows is easier to learn than a command language-based interface like DOS. This is also why ACTION BUTTONS (K4) should always have a textual label to go along with the graphical icon.

4. Help customers avoid and recover from errors. Errors cause frustration, poor performance, and a lack of trust in your site. PREVENTING ERRORS (K12) will help avoid many of these problems. Unfortunately, no matter how well you design the site, humans will make occasional errors. Help people recover from errors by presenting MEANINGFUL ERROR MESSAGES (K13). Tell them what happened and how to recover, or better yet, offer to automatically carry out the steps that would help them recover from the error.

4 Involving Customers with Iterative Design

5. Support customer control and freedom. Customers should sense that their actions determine the site's responses, and that they're not being forced down a fixed path. Providing MULTIPLE WAYS TO NAVIGATE (B1) is one example of how to support this impression on your site. It also means that the customer is given easy exits, such as undo and redo options, for mistaken choices. The browser's built-in **Back** and **Forward** buttons and LOCATION BREAD CRUMBS (K6) are mechanisms that give customers easy exits on the Web.

6. Help frequent customers use accelerators. Keyboard shortcuts are important for expert customers. Your site can support frequent actions automatically. For example, your site can store information, such as shipping addresses, and use PREDICTIVE INPUT (H11) so that your customers do not have to retype this information every time they come back to a page. Design an ACCOUNT MANAGEMENT (H4) interface that makes it easy for your customer to see and change this stored information.

7. Strive for aesthetic and minimalist design. Clean aesthetics make using your site a pleasing experience. A GRID LAYOUT (I1) is one common technique you can use to ensure that your site has a clean, understandable look. Well-designed type, images, and graphical elements communicate how the site works. Often visual elements are overused. If removal does no harm to the site, take out irrelevant information and graphics from all pages. Every extra element draws attention away from the ones that matter.²

These principles sometimes conflict. Use your best judgment to resolve these conflicts. For example, supporting individual control and freedom may conflict with helping customers avoid errors. Sometimes, as in the case of a PROCESS FUNNEL (H1), restricting control and freedom can help customers complete their tasks. Use these restrictions judiciously, though. We have tried to make these conflicts apparent when discussing the forces that come to bear on each pattern in Part II of the book.

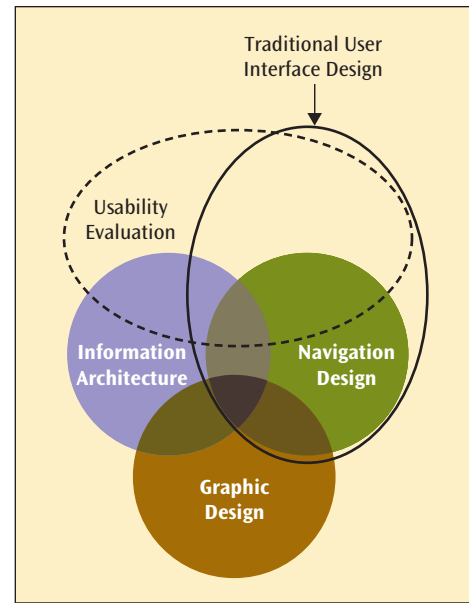
Information Architecture, Navigation Design, and Graphic Design

No matter what form of design you're carrying out—information architecture, navigation design, or graphic design—use the design goals and

² As Edward Tufte says, "Graphical excellence is that which gives the viewer the greatest number of ideas in the shortest time, with the least ink in the smallest space" [from *The Visual Display of Quantitative Information* (1983), p. 51].

Figure 4.3

The spaces of information architecture, navigation design, graphic design, and usability evaluation overlap. Traditional user interface design is primarily navigation design and usability evaluation, with a touch of information architecture and graphic design.



4.3

principles that we have described to guide your work. **Information architecture** means identifying, structuring, and presenting groups of related content in a logical and coherent manner.³ **Navigation design** means designing methods so that customers can find their way around the information structure. **Graphic design** means developing the visual communication of information, using elements such as color, images, typography, and layout. Information architecture and navigation design are typically carried out before graphic design.

Figure 4.3 shows how these three types of design relate to one another, as well as how usability evaluation and traditional user interface design fit into the picture. All three approaches to design are necessary.

³ There is an ongoing (and sometimes heated) debate in the design community about the delineation of *information architecture* versus *information design*. In general, information architecture focuses more on things like structure and language, while information design concentrates on presentation and perception. However, distinctions between these two fields are still very blurry. Information architecture and information design represent a convergence of multiple disciplines with different backgrounds, vocabularies, and cultures. The key here is to go beyond these superficial differences and to focus on what they all have in common: helping customers find, understand, and manage complex information.

4 Involving Customers with Iterative Design

A graphic design that is rich with images cannot compensate for a poor information architecture. Likewise, a clean information architecture cannot make up for a navigation design that hides the location of all the navigation elements.

Rapid Prototyping

As we said earlier, a key principle of iterative design is rapid prototyping—quickly creating rough-and-ready mock-ups that provide useful feedback. These prototypes can help reduce risk, lead to smaller and less complex systems, and nail down what customers really need. In the sections that follow we describe how to do rapid prototyping, beginning with a description of the different design artifacts that are created during iterative Web site design.

4.4

Site Maps, Storyboards, and Schematics

Typically, early in the design and prototyping phases three kinds of design artifacts are developed: site maps, storyboards, and schematics. A **site map** is a high-level diagram showing the overall structure of a site (see Figures 4.4 and 4.5). It is used primarily to reflect an understanding of the information structure, or architecture, of the site as it is being built and, to a limited extent, the navigation structure, or **flow** through the site.

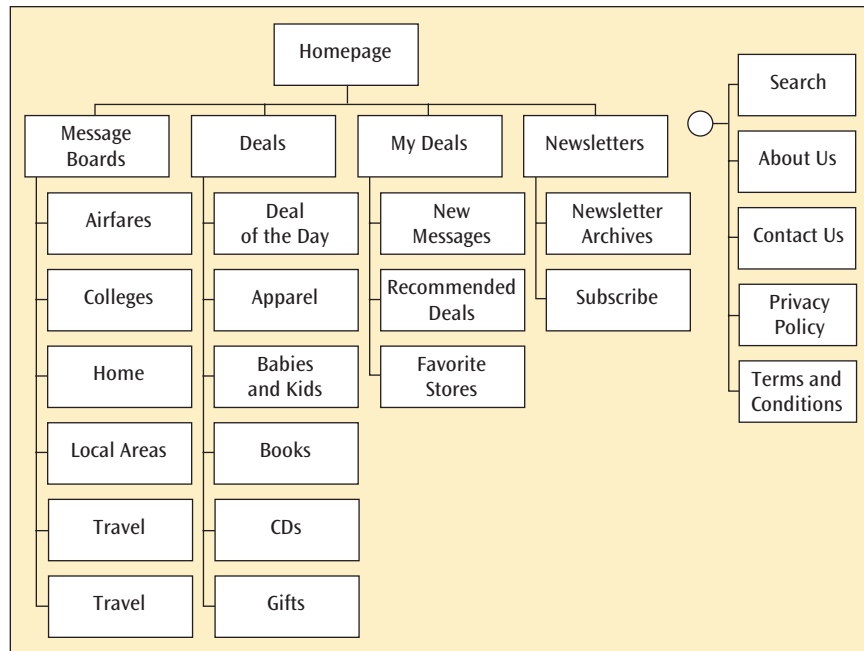
A **storyboard** is a sequence of Web pages depicting how a customer would accomplish a given task (see Figures 4.6 and 3.3). Use storyboards to illustrate important interaction sequences, or flows through a site. When showing ideas to a client, you can accompany storyboards with a narrative about the task that the customer is trying to accomplish. That is, you might show the client the complete **scenario** you developed, or at least a small piece of it.

Schematics are representations of the layout and content that will appear on individual pages (see Figure 4.7). They are usually devoid of images, though they may indicate, with a label, where to place an image. The fonts, colors, and layout are often quite preliminary, not indicating a final decision, but instead giving the graphic designer hints about which information needs to be highlighted or grouped together.

You probably noticed that the illustrations of site maps, storyboards, and schematics here look pretty basic. At this stage they are just abstract representations, not to be taken literally. Their purpose is to get the big ideas across, without the irrelevant details that distract reviewers.

Figure 4.4

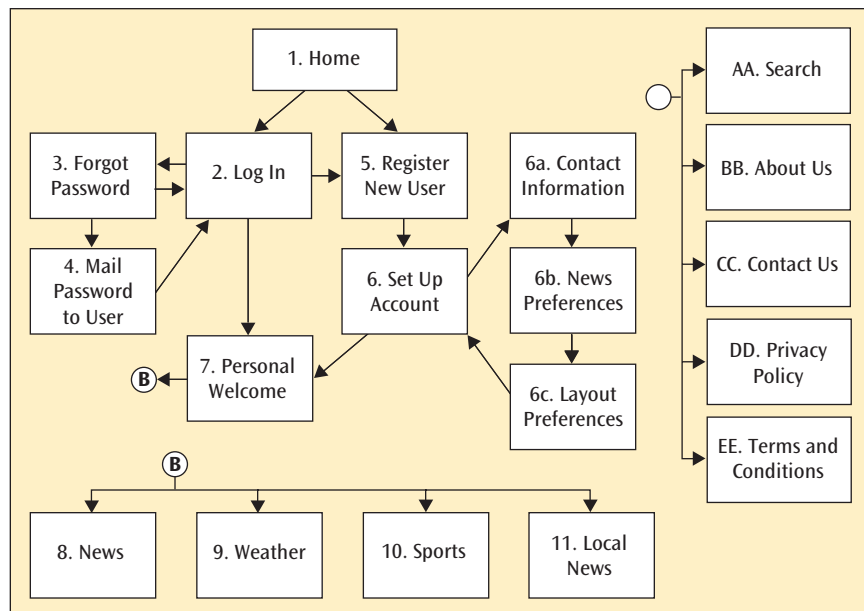
A site map is a high-level diagram that depicts the overall organization of a site. This site map shows the structure of a Web site that helps people find online deals and electronic coupons. The set of pages on the right represents a navigation bar that is available on every page.



4.4

Figure 4.5

This site map shows part of the structure and flow of a personalized news site. [This design has a problem: it forces people to sign in before they see any content. See the SIGN-IN/NEW ACCOUNT (H2) pattern for more details.]



4.5

4 Involving Customers with Iterative Design

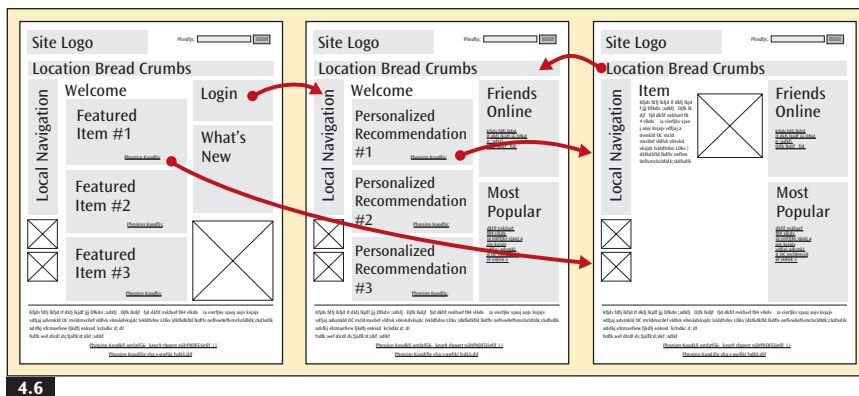


Figure 4.6

Storyboards show the steps that a customer would take to accomplish a task. This storyboard shows how a customer interacts with a site that lets groups of friends find, recommend, and share things with each other.

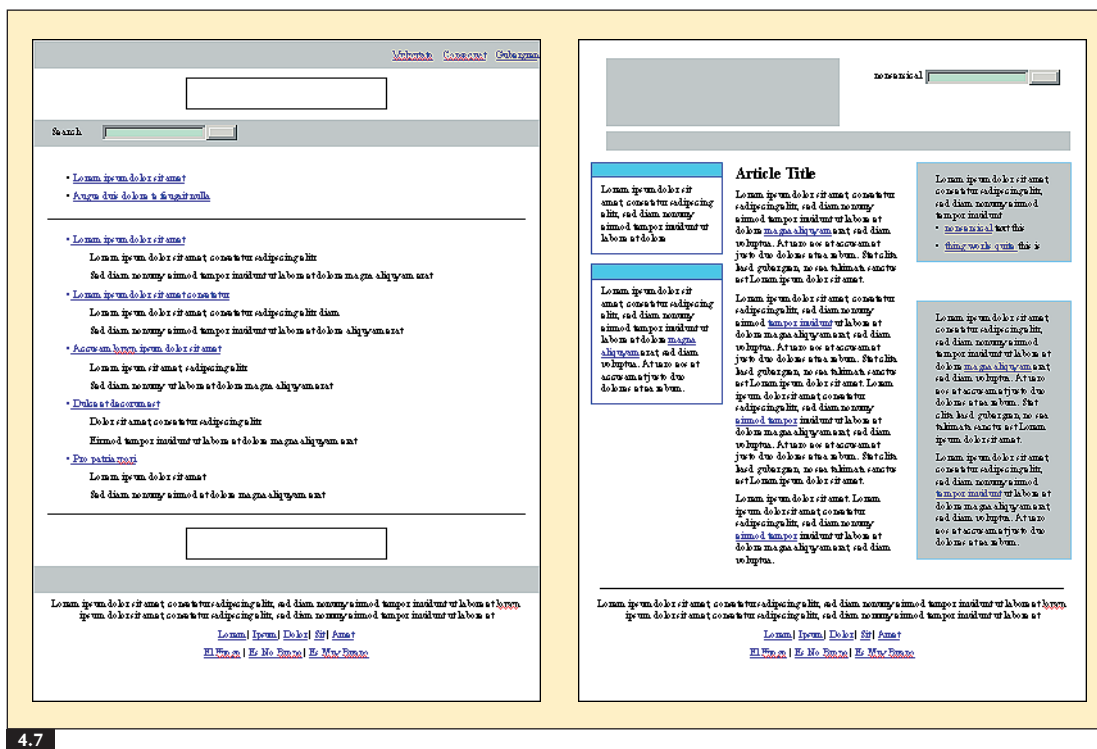


Figure 4.7

These two examples of schematics have the feel of complete individual Web pages.

Research that we carried out with Mark Newman, a human–computer interaction researcher, found that often designers first sketch out all three of these representations on paper (see Figures 4.8, 3.3, and 4.9). Before showing these representations to clients or customers, many designers like to clean them up by creating electronic versions. Others are comfortable showing the informal, sketched representations to get validation earlier in the process and keep the discussion focused on the important issues. Cleaning things up too much may cause your reviewers to improperly focus on the colors, fonts, and visual alignment. It really depends on your clients and their expectations. If you manage their expectations well, your clients will understand why you’re showing them rough sketches.

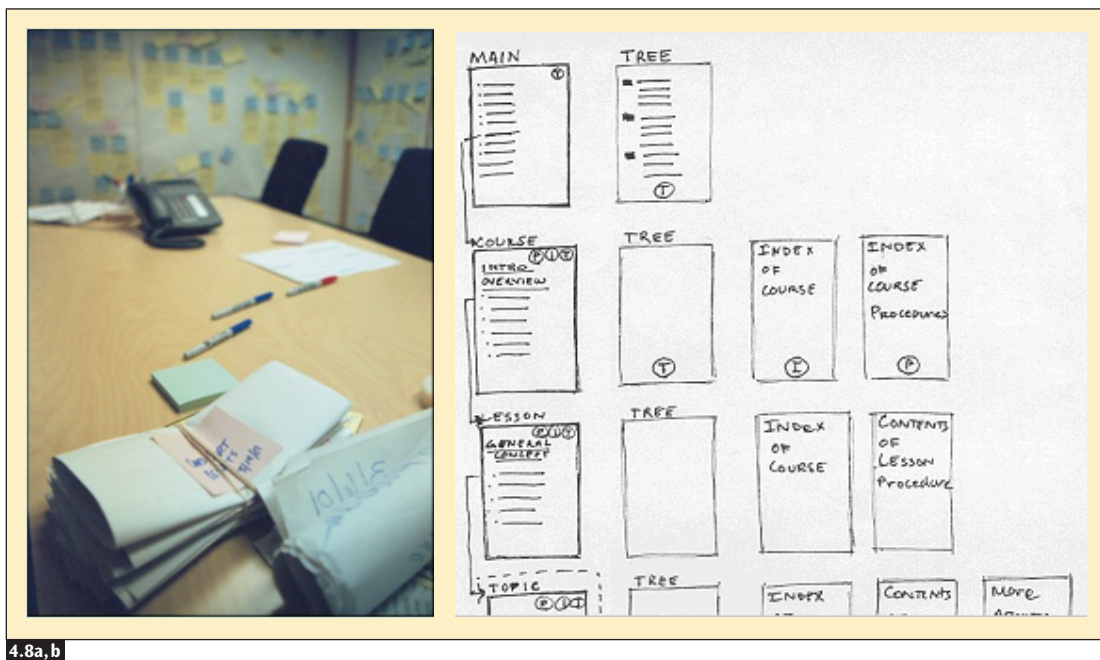


Figure 4.8

Create site maps on walls, using pens, papers, and Post-it notes (a). This technique, known as *affinity diagramming*, lets multiple team members brainstorm simultaneously and provides a large, immersive display for working on large sites. When you’re finished brainstorming, copy the site map to paper (b) or take a digital photo to capture the design.

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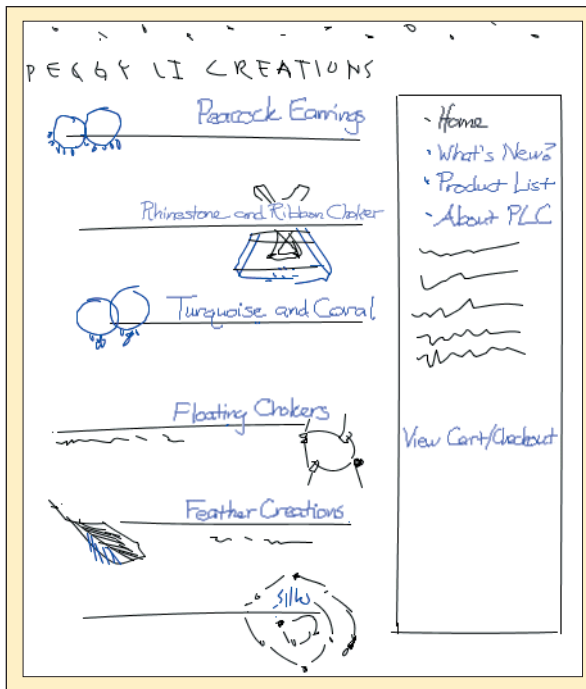


Figure 4.9

A low-fidelity prototype is a quickly created, rough sketch that gives an overall feeling of what's needed on the page, without going into unnecessary detail. This sketched schematic illustrates the main functionality available at a Web-based custom jewelry store.

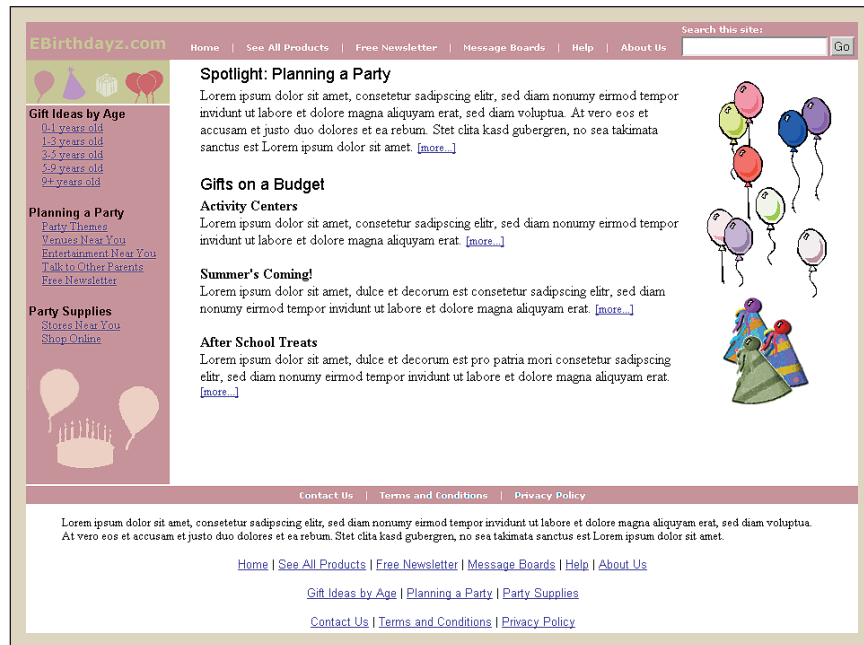
Progressive Refinement, from Low Fidelity to High Fidelity

The great thing about prototypes is that they can be created quickly and used to get feedback from customers. Low-fidelity prototyping is one technique that many designers use to accomplish this task. In **low-fidelity prototyping**, you use paper, whiteboards, Post-it notes, and markers to create rough cuts of a Web site. That's it. Sketches are low-fidelity ("low-fi") when they are far from the final design in both their visual and their interactive details. For example, Figure 4.9 shows hand-drawn graphics and handwritten text, much of it represented by squiggly lines. In addition to sketching, you can use cut, copy, and paste techniques with scissors, glue, and photocopying machines.

Using a set of low-fidelity pages, you can test a design with representative customers. Sit them down in front of your sketches and ask how they would complete a particular task. On the basis of the customer's verbal responses or pointing, one of your teammates can "play computer" and flip to a new page to show the designed output of the site. Observing what customers do with these low-fi designs will give you valuable information

Figure 4.10

A high-fidelity prototype, such as this one, is detailed and rich with typography, color, and graphics.

**4.10**

about how to refine your early design ideas. This type of prototyping and testing can let you go through an entire iterative design cycle—design, prototype, evaluate—in less than a day.⁴

Our personal experience has shown that it is *at least* ten to twenty times easier and faster to create a low-fidelity prototype than an equivalent high-fidelity (“hi-fi”) prototype. By **high-fidelity prototype**, we mean one that looks polished and complete, created with computer-based tools such as Macromedia Dreamweaver or Adobe Photoshop or Illustrator (see Figure 4.10). We’re not saying that you don’t need hi-fi prototypes—just that you don’t need them in the *early stages of design*. It’s not worth the effort of focusing on colors, fonts, and alignment when there are more important issues, like organization and overall site structure, to worry about.

⁴ Read Mark Rettig’s article, “Prototyping for Tiny Fingers” (see the Resources section later in the book), for a great how-to on creating low-fidelity prototypes and testing them with customers. Another, more in-depth resource in this area is Carolyn Snyder’s book titled *Paper Prototyping*.

Another advantage of low-fi prototypes is that, because specific programming or graphic design skills are not required, the contributions and insights of each team member can be easily integrated into the design. Even CEOs have been observed creating low-fi prototypes! Creating low-fi prototypes and evaluating them with customers is a good team-building experience, even for people whose roles traditionally do not include interaction with customers. Going through this process will get everyone on the same page about what customers really need. You will find that it is a lot more effective and fun to create low-fi prototypes than to argue endlessly about what customers *might* want.

Avoid Computer-Based Tools in the Early Design Stages • Research shows that designers who work out conceptual ideas on paper tend to iterate more and explore the design space more broadly, whereas designers using computer-based tools tend to take only one idea and work it out in detail.

Nearly every one of the designers we have talked to has observed that *the discussion is qualitatively different when people are presented with a high-fidelity prototype*. Clients often respond with comments like, “I do

What’s in a Paper Prototyping Kit?

It’s a good idea to create a paper prototyping kit, a small box of goodies for making paper prototypes that everyone on your team can access. Here’s a shopping list of supplies that every kit needs to be complete:

- Lots of paper (both white and colored construction paper)**
- Lots of index cards**
- Lots of Post-its**
- Transparencies**
- Scotch tape**
- Scissors**
- An X-ACTO knife**
- Paste**
- Markers (with lots of colors)**
- Pens**
- Rulers**
- Duct tape (hey, you never know when it will come in handy)**

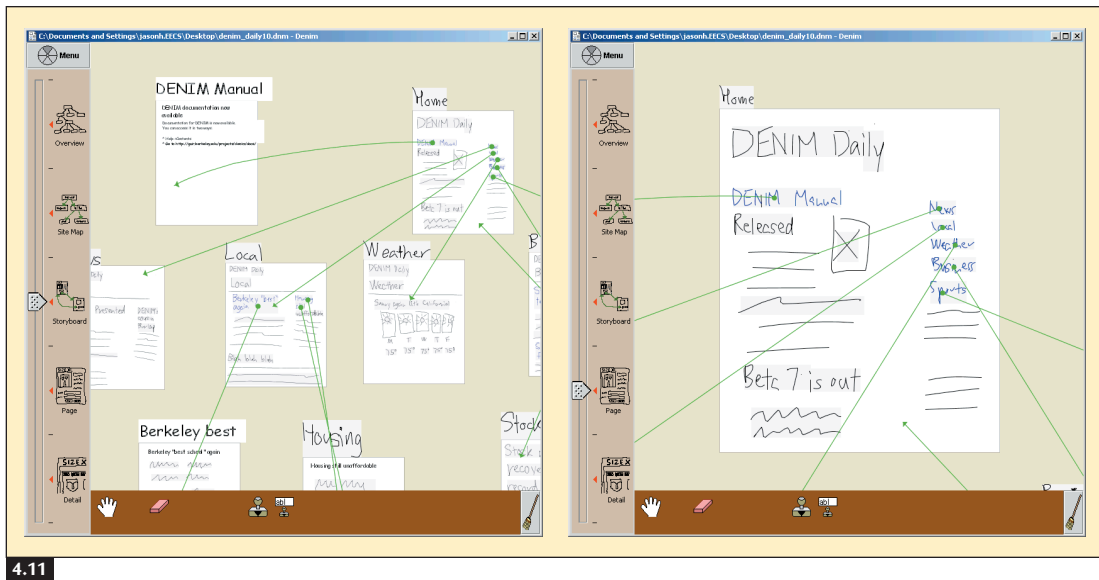


Figure 4.11

These two screen shots illustrate how DENIM, a sketch-based Web site design tool for the early stages of design, allows you to quickly sketch Web pages, create links among them, and interact with them in a run mode. You can use zooming to integrate the different ways of viewing a site, from site map to storyboard to individual pages.

not like your color scheme,” or “These two buttons need to be aligned correctly.” When presented with a low-fidelity prototype, however, clients are more likely to say something like, “These labels on the navigation bar do not make sense to me,” or “You’re missing a link to the shopping cart here on this page.” In other words, with low-fidelity prototypes, which lack irrelevant details like color, font, and alignment to distract the eye, people focus on the interaction and on the overall site structure.

All of the tools used in Web design today focus on creating finished products. Tools like Microsoft FrontPage, Adobe GoLive, and Macromedia Dreamweaver help you create production Web sites, not early prototypes. Other tools used by designers, such as Microsoft Visio, Macromedia Director, Adobe Illustrator, and Adobe Photoshop, have the same problem. Until tools that support the progression from low-fidelity to high-fidelity prototypes become widely available, we advocate that you delay using computer-based tools. Incidentally, we have developed a research-quality tool called DENIM that allows you to “sketch” low-fidelity prototypes on a computer (see Figure 4.11).⁵

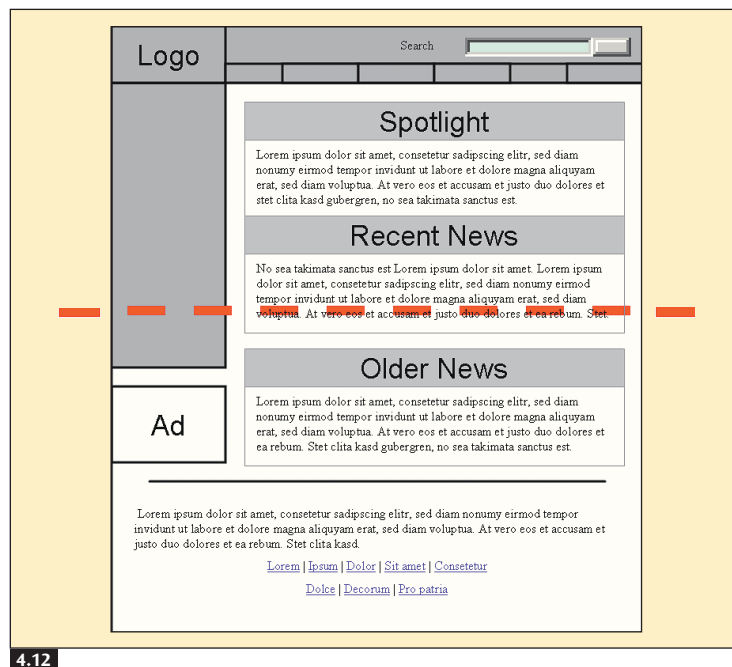
5 If you are ambitious, try out DENIM at dub.washington.edu/denim.

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When You're Ready, Switch to Computer-Based Tools • When should you move to computer-based tools? It depends on your work practices. If you find that you absolutely must save designs and e-mail them to others, you might be ready. Design teams often want to switch to using a computer when they're presenting to clients. In our interviews, the general consensus was that you should not show low-fidelity sketches to clients, because they are perceived as unprofessional. However, high-fidelity mock-ups take a long time to create, and again they have the drawback of directing the discussion toward extremely fine details too soon. One solution is to go ahead with sketches and, as we mentioned before, manage clients' expectations by explaining that you're using sketches instead of computer-based designs to speed up the iteration process and to focus on the right issues at this stage.

Another solution is to use what we call **medium-fidelity prototypes** (see Figure 4.12). Medium-fidelity prototypes have many more details about content, but they do not distract clients or customers with fonts, colors, and graphics. Medium-fidelity prototypes are a good compromise if you need to present mock-ups.

At some point your team will have most of the major structural and interaction issues hammered out and satisfactorily tested with customers.



4.12

Figure 4.12

A medium-fidelity prototype is a cleaned-up illustration that shows more detail about content, without specifying typography, color, or graphics. This kind of prototype is often shown to clients in lieu of low-fidelity or high-fidelity prototypes.

This is a good time to create high-fidelity prototypes—ones that are richer and closer to what the final site will look and feel like. You will probably create these hi-fi prototypes with HTML-based tools such as Dreamweaver and graphic design tools such as Illustrator and Photoshop.

Again, the key here is to fake it! For example, suppose your site lets people view stock prices. You do not have to show the real prices, or even real graphs. Just create one or two sample images and use them for all of the graphs. You will show enough to give people the feel of the final site. As another example, if you're building a site that uses personalization technology to improve the customer experience, when applying **PERSONALIZED CONTENT (D4)** you might mock up the customization at this stage and make it look the same for all of your test participants.

Horizontal and Vertical Prototypes

In the very early stages of design, you often want to create prototypes that show a broad swath of what the eventual Web site will support. These **horizontal prototypes** might show the top-level pages, but without much depth behind them. The homepage might have all the links you expect to have, and each will take you somewhere, but any links from those second-level pages that implement specific features will not exist yet. These prototypes are good for making sure the basic features of the site are present and organized logically.

In contrast, sometimes you will want to flesh out and test the steps that a customer will go through to complete a particular task, such as **SIGN-IN/NEW ACCOUNT (H2)**. A **vertical prototype** implements only the key pages along the path for completing a particular task. This step is appropriate when a complex feature is poorly understood or needs to be explored further. You will not yet support any links that connect to other tasks or to other parts of the site.

Often you will want to combine these two techniques, as in Figure 4.13. Prototype the entire top level of the site with a horizontal prototype to give a flavor of what will be on the site. Then focus on one particular feature, and use the vertical approach to prototype the pages illustrating that feature in detail.

Limits of Prototyping

There are some downsides to using prototypes. Many people believe that creating prototypes takes time away from building the actual site. Take care to explain the value of creating and testing *rapid* prototypes. In addition, although prototypes are useful for getting some kinds of information, they

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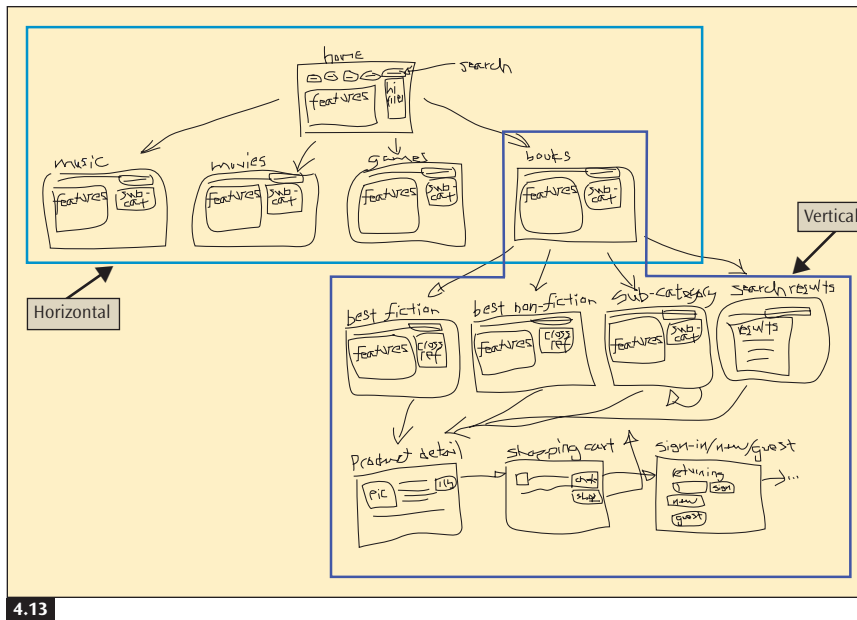


Figure 4.13

This figure shows a combination of the horizontal and vertical prototypes. Whereas horizontal prototypes illustrate the broad functionality of a design at a high level, vertical prototypes take one feature and detail it in depth. A combination makes a useful demonstration of the broad features in a site (horizontal) and the detailed functionality (vertical) as exemplified by one section.

are not as effective for others, such as estimating download speeds and quality of customer service. Finally, when using medium- and high-fidelity prototypes, you need to manage expectations. Seeing something that looks and feels as if it's working can make customers and clients think that the site is almost done. Take the time to explain that the prototypes still represent early stages of design, and that things are still open to change.

Evaluating Your Web Site

Web site evaluation is the third (and most often overlooked) part of iterative design. Without evaluation you cannot know if you have met your target goals. You can evaluate your Web site by having it reviewed by experts, by informally testing it with a few participants, or by conducting a more formal study with customers. Each of these methods has its pros and cons, as described in the sections that follow.

Expert Reviews

Expert reviews are an effective technique for evaluating sites without involving customers. The most common type of expert review is a **heuristic evaluation**, which was developed by noted Web guru Jakob Nielsen.

4.5

The basic idea is have three to five expert judges independently evaluate a site, using a list of usability heuristics or principles. Nielsen's site, useit.com, lists ten such heuristics, though the seven design principles that we listed earlier in this chapter would work just as well.

In a heuristic evaluation, the judges go through the site, often with a set of sample tasks as a guide, looking for violations of the heuristics. They note each violation and make a suggestion for fixing it. For example, if a site were found to use different terminology for the same concept on two different pages, the judges would note a "consistency" violation and suggest using one of the terms on both pages.

The judges also rate each violation with a level of severity. Severity levels are usually assessed on the basis of the expected customer impact and frequency of the violation.

After each judge has independently evaluated the site, all the judges compare their lists of violations. The point of this step is to merge the lists into one and find agreement on the severity levels. Nielsen's research shows that multiple experts will find different problems, and only by using multiple judges will you have a high probability of finding most of the problems on the site. The more expertise your judges have on user interface design, usability, and heuristic evaluation, the fewer judges you will need. If you can get true experts, we recommend you use three judges. If you're using any team member you can get your hands on, we recommend you involve five or six.

Heuristic evaluation can be an especially effective technique for finding potential usability catastrophes. It also works well for identifying subtle problems that most usability studies might not reveal, such as a poor choice of colors. We have found that heuristic evaluations do not work as well on low-fidelity prototypes as on high-fidelity prototypes. Inexperienced judges tend to focus on the pieces that are missing. We recommend not carrying out a heuristic evaluation until you have a high-fidelity HTML prototype of your site.

With any heuristic evaluation, keep in mind the following caveats: Often experts know too much or not enough. They might be much more sophisticated than your customers and overlook things that will trip up real customers. On the other hand, your site might be designed for customers who have a lot of domain knowledge, such as doctors, and the judges might not have the background necessary to understand the site.

More importantly, heuristic evaluations have a tendency to reveal lots of false positives. **False positives** are violations that are identified by the heuristic evaluation but never found in a usability study of the

same interface. In other words, experts often find problems that do not turn out to be problems in practice. Fixing these false positives could end up wasting lots of critical design and engineering resources. Still, a heuristic evaluation is inexpensive, can be carried out in a few hours, and is good for revealing possible usability problems. We suggest that you review Nielsen's site and papers for details on carrying out a heuristic evaluation, and use this technique in tandem with informal and formal usability tests.

Informal Evaluations

Informal evaluations are a natural outgrowth of rapid iteration and low-fidelity prototyping. The idea is quite straightforward: recruit five to ten people who are representative of your target customers, show them your Web site prototype, ask them to do some of the tasks from your task analysis, and take good notes. Your goal is to obtain qualitative feedback from customers about what works and what doesn't, both from what they say and from what they actually do.

The prototype does not have to be computer based. It can be as simple as a paper prototype. In this case, have people point and click with their fingers, just as they would do with a mouse.

You might recruit several representative customers to come to your offices and ask them to complete the tasks with the prototype. Alternatively, you might visit customers in their homes or offices and ask them to do the same.

Before you begin, ask participants to "think aloud," to say what's going on in their minds. This is known as a **think-aloud**, or **verbal, protocol**. They will probably find this a little strange at first, but they'll get used to it quickly. You may have to prompt them every so often by asking a question like, "So, what are you looking for now?"

The data collected in informal evaluations is qualitative **process data**. This kind of data gives an overall feeling of what works and what doesn't. While participants are testing, keep an eye out for instances in which they seem confused, say something negative, or even swear. These are called **critical incidents**. Use them as a starting point for places you will redesign. Look for positive incidents, cases in which the customer liked your site or things appeared to be going smoothly. Positive incidents give you a hint about which parts of your design work well, and you might be able to take advantage of some of the same ideas elsewhere.

We do not recommend that you use informal evaluation techniques to show that one site design is better than another, or to say how long certain actions will take. Instead, use the results to identify potential problem areas

that need to be improved. After you think you have solved the problems, rerun the tests with a new set of representative customers and see if the problems have been addressed properly.

See the appendixes for more information on how to set up and run usability tests. They lay out the roles the testers assume, scripts for what to do and say during the evaluation, and tips for how to analyze the information you collect.

Formal Usability Studies

As soon as you have a running prototype of your Web site, you can start getting hard numbers on whether the site meets the goals that your team has set. For example, one of your goals might be that customers be able to register and create an account in less than two minutes. You can run this study with as few as ten participants, though it generally takes more participants when you want to test numerical goals.

The type of information to collect in this situation is quantitative **bottom-line data**. Bottom-line data consists of hard numerical metrics where you're looking for statistical significance—a serious and reliable difference instead of one due to chance.

Bottom-line data is especially important for simple, repetitive interactions; for example, “Are customers completing the task faster when this button is placed on the left or right side of the page?” This type of data is also useful for comparing two different interfaces; for example, “Can more people successfully make purchases and check out using shopping cart interface A or B?”

Online testing makes it easy to test a variety of similar issues. It lets you recruit and test many participants to achieve statistical accuracy quickly. Several companies offer products to set up and run online tests. These sites can recruit research participants for you, or allow you to enter a list of e-mail addresses from current customers or your own participant pool. The sites then automatically e-mail research participants, lead them to a test site, and ask them to complete tasks that you have defined in advance. Most systems allow you to track the pages the participants go through, as well as ask survey questions.

Some practitioners have developed creative—and potentially controversial—ways to measure usability. One Web site study used “returns to the homepage” as an indication of an error. The inference was that people got lost and had to go back to the homepage. This might be a big assumption without further data to back it up. What if customers want to return to the homepage simply because they're done with a particular section?

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If you want to measure the number of errors, make sure that you define errors beforehand and that everyone agrees on the definitions.

See the appendixes for more details on running formal usability evaluations, both offline and online.

Choosing an Evaluation Technique

When do you use expert reviews, informal evaluations, and formal evaluations? The answer is to use multiple techniques. The techniques should be balanced by your cost constraints, as well as by how early in the process the evaluation is taking place. For example, testing low-fi prototypes is especially effective in very early stages of design. In later stages, because you have defined more of the site details, you might want to create HTML prototypes instead so that you can evaluate these details.

Always run expert reviews and informal evaluations as you iterate, to work out basic design flaws. To help you figure out where the problems are, focus especially on qualitative process data in the early stages of design. Use the patterns to find solutions to these problems and then iterate. Do a few formal evaluations as your Web site matures, as it gets closer to deployment, or even after it has been deployed. This strategy will help you refine and polish your site.

Sometimes using five to ten test participants will be good enough to convince you and your teammates that the problems found during testing are legitimate design issues that must be resolved. However, it will often be harder to convince management or the marketing organization that you need to make changes, especially on a high-traffic page like the homepage. In that case, use more formal usability studies and techniques, such as online usability testing, which makes it easy to test the site quickly with fifty to two hundred customers.

A small number of test participants uncover only a small number of the potential problems, according to research by the consulting firm User Interface Engineering.⁶ This firm's theory is that many Web sites attract a varied set of customers who have a wide range of goals that they're trying to accomplish, so a small number of participants cannot accurately reflect the diversity of customers. Again, this problem can be overcome by testing with a larger number of customers, on a larger variety of tasks. We believe that online usability testing is a necessity because traditional usability testing with fifty to two hundred participants would take too long.

⁶ Cited in C. Perfetti and L. Landesman, "Eight Is Not Enough" (2001) (www.uie.com/articles/eight_is_not_enough).

4.6 Take-away Ideas

The key to designing successful Web sites is to follow a customer-centered, iterative process that first identifies the expected customers and their tasks. Following the principles of (1) keeping the customers involved, (2) conducting rapid prototyping, and (3) evaluating designs leads to design ideas and prototypes that you evaluate with real customers. The evaluation leads to redesign, and then you iteratively repeat this process until you achieve your usability and business goals. Involving your customers throughout this process will keep your design on track.

Set your team's usability goals early in the design process, and continually evaluate your progress toward these goals. The iterative design process is the least costly way to improve the site design. When the site launches, continue to monitor ongoing customer metrics to see how you're doing, to inform changes, and to help set your goals for the next version of the site.

